THE LONG RUN AGAINST THE CPU DEALER

100,000 rounds. *Still noise.*

Six decks, one unit a hand, and the total-dependent chart.
The same engine as Session 001, played 1,000 times longer.



NET RESULT

−241.00 units

W–L–P	OBSERVED EDGE	EXACT EDGE
43,246–47,547–9,207	0.241%	0.426%

6 decks · S17 · 3:2 · Double any 2 · DAS · 3 resplits

SEED 8675309 · 75% PENETRATION · COMMIT 6e0fc28

THE SHORT VERSION

The long run is *still noisy.*

After 100,000 rounds the chart was down 241 units. The exact solver expected a loss of 425.61 units. The sample finished 184.61 units better than expectation — about 0.51 standard deviations. That is ordinary luck, not evidence that the chart is wrong.

NET RESULT	OBSERVED EDGE	EXACT EDGE	95% INTERVAL
-241.00	0.241%	0.426%	-0.47%
units over 100,000 rounds	against the player	composition-aware solver	to 0.96%

The run

Rounds	100,000	Hands after splits	102,840
Wins	43,246 · 43.2%	Losses	47,547 · 47.5%
Pushes	9,207 · 9.2%	Graded actions	117,458
Net per round	-0.00241 units	SD per round	1.153 units
Expected net	-425.61 units	Net vs exact	+184.61 units

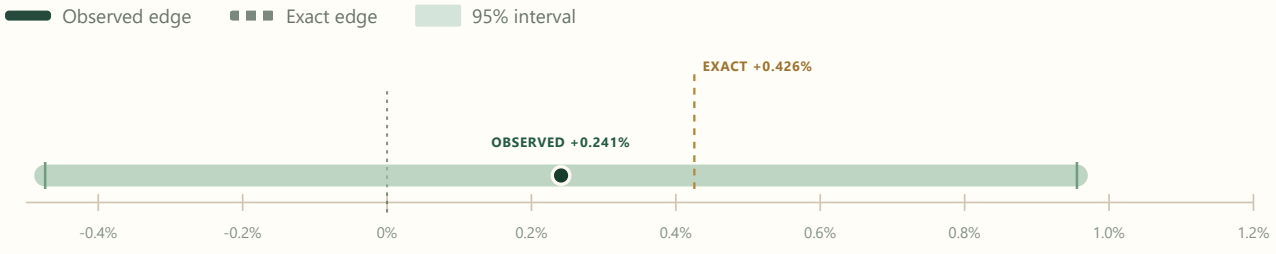
The reference seed is 8675309. A round includes all split hands; the 100,000 rounds became 102,840 settled hands.

Read the interval

The standard error of the observed edge is **0.365 percentage points**. The 95% interval runs from **-0.474%** to **0.956%**.

The exact 0.426% edge is inside that interval. So is break-even. The result is consistent with both a tiny player edge and a house edge above 1%.

> THE POINT
>
> 100,000 rounds is enough to measure the engine. It is not enough to measure a 0.4% edge.



95% CONFIDENCE INTERVAL FOR THE OBSERVED EDGE: -0.474% TO +0.956%

A 95% confidence interval is a statement about the method, not a promise about this run. The interval is wide because one round can swing more than a full bet.

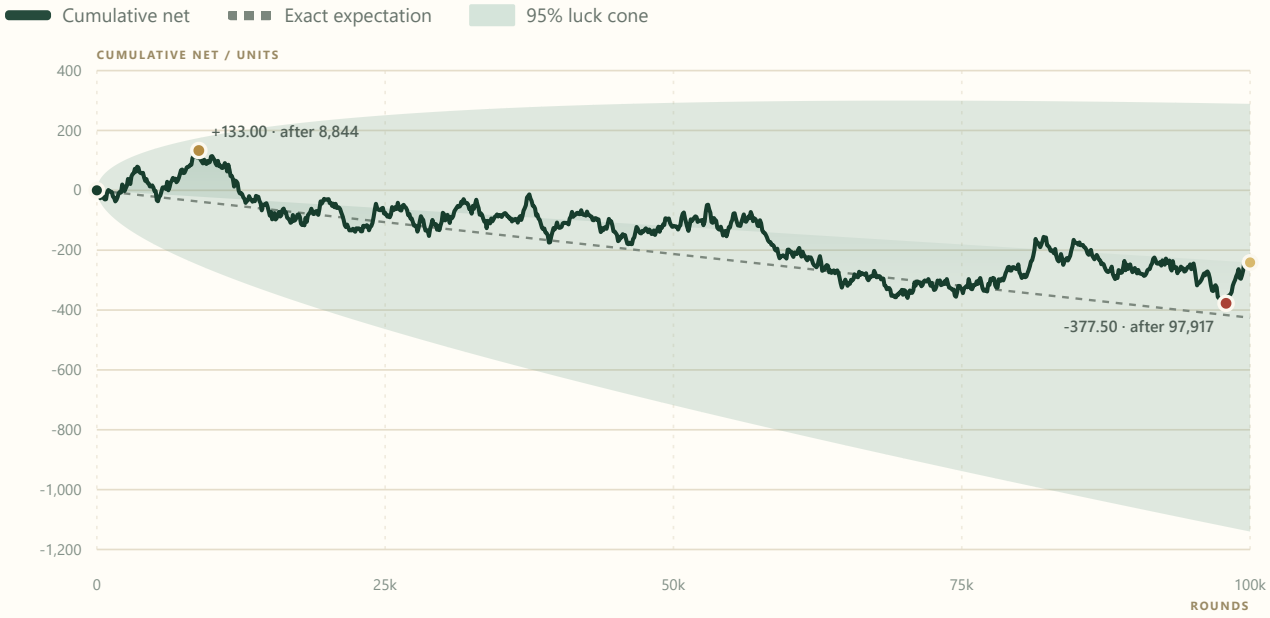
WHY THE MISS IS SMALL

The gap between the observed and exact edge is **0.185 percentage points**. The standard error alone is **0.365 points**. The simulation did not fail to find the edge; it found a sample whose center is very close and whose error bar is wider than the effect.

THE LUCK CONE

Down 241 units. *Well inside the cone.*

Cumulative net over 100,000 rounds. The dashed line is the exact expectation; the shaded cone is the 95% range for a flat-betting player with the same policy.



Expected path: 0 → -425.61. The 95% band at round 100,000 is roughly ±714.54 units. The observed path ends +184.61 units above expectation.

PEAK	TROUGH	MAX DRAWDOWN	LONGEST LOSS STREAK
+133.00	-377.50	510.50	11
after round 8,844	after round 97,917	units, peak to trough	rounds in a row

The path peaked at +133.00 after round 8,844, then spent most of the run below zero. Its lowest point was -377.50 at round 97,917.

The final result is 0.51 standard deviations above the exact expectation. The cone is 714.54 units wide at the end. A flat 100,000-round result is supposed to look like this.

THE HOUSE EDGE IS AVERAGE, NOT A TREND

Blackjack does not lose a little on every round. The player wins 43,246 rounds, loses 47,547 and pushes 9,207. Naturals pay 3:2, and doubles and splits change the size of the bet. The 0.426% edge is the average of all those outcomes, not a line the bankroll follows.

HOW THIS WAS PLAYED

Reproducible. *Not proof.*

This is a headless run through the same `PlayGame` engine that powers the browser table, not a hand-typed sequence. The seed fixes every shuffle and every draw.

Rules

Decks	6
Dealer	Stands on soft 17
Blackjack	3:2
Double	Any two · after split
Splits	Up to 4 hands · aces one card
Surrender	None
Dealer peek	US peek
Insurance	Offered

Run

Seed	8675309
Rounds	100,000
Hands after splits	102,840
Bet	1 unit, flat
Penetration	75%
Strategy	Total-dependent chart
Grading	117,458 decisions · 99.91% on-chart
Code	6e0fc28

What is exact

Dealer distributions and action EVs are exact for their composition. The 0.426% fresh-shoe edge is the composition-aware fixed-strategy solve; its split EV uses the documented independent-hand approximation.

The simulation deals from the real six-deck shoe with a 75% cut card. It settles splits, doubles, insurance and dealer peeks with the same rules the table uses.

What is approximated

Split EV uses the independent-hand approximation. The chart is total-dependent, not composition-dependent. The insurance grade treats the hole card as one unseen card rather than a peek.

The play harness matched the chart on 99.91% of decisions. The 105 exceptions are rare post-hit or legal-action states; their combined EV cost was 14.71 units over the run.

What this does not prove

One hundred thousand rounds cannot pin a sub-1% edge to three decimals. The 95% interval on the observed edge spans -0.47% to 0.96%, so this sample is consistent with break-even and with a house edge above 1%. The exact solver is the source of the 0.426% number; the simulation is a consistency check, not a replacement.

REPRODUCE IT

```
node reports/build-blackjack-100k-report.mjs
```

Writes `blackjack-100k.json`, this HTML and the PDF. The reference seed is 8675309; the 100-round session seed is 20261029.

THE LAST WORD

The edge is real. The sample is still mostly a story about variance.